CALCULUS

Chapter 17

Differential Equations

17

SYLLABUS 5.18 5.19 HL only

A cup of coffee cools fastest when it is hottest. Pull it from the machine at 80 degrees and it drops quickly at first; near room temperature it barely changes. The *rate* at which it cools depends on how far above the room its temperature currently is.

That sentence is a differential equation. Written in symbols, the rate of cooling $\frac{d\theta}{dt}$ is proportional to the gap $\theta - 20$ between the coffee and the 20-degree room:

$$\frac{d\theta}{dt} = -k(\theta - 20).$$

Notice what kind of equation this is. The unknown is not a number but a *function*, $\theta(t)$, and the equation links the function to its own derivative. Most laws of nature take exactly this form, because what we can observe directly is usually a rate: how fast something grows, cools, decays, or moves.

A **differential equation** relates a function to its derivatives. To **solve** one is to find the function itself, recovering $\theta(t)$ from a statement about $\frac{d\theta}{dt}$.

That is worth doing, because it turns a rule about change into a prediction. A law of nature tells you what is happening at this instant; solving its differential equation tells you the whole future.

By the end of this chapter you will be able to solve such an equation exactly by three methods: separating the variables, substituting to make them separate, and using an integrating factor. You will solve one approximately by Euler's method when no exact solution exists. And you will write a function as an infinite polynomial, a Maclaurin series, which approximates the function and gives a fifth route to a solution. The coffee equation is solved in the first section.

17.1 Separable Variables

Differential equations are classified by what they contain, and two words fix the ones in this chapter. The **order** is the highest derivative that appears, so every equation here, containing $\dfrac{dy}{dx}$ and no higher derivative, is **first order**. The equation is **linear** when y and its derivatives appear only to the first power and are never multiplied together; $\dfrac{dy}{dx} + xy = 1$ is linear, while $\dfrac{dy}{dx} = 2xy^2$ is not. The three exact methods below are chosen by which of these forms an equation takes.

The simplest case can be rearranged so that each variable sits on its own side. A first-order equation is **separable** if it can be written as a function of y times a function of x:

$$\frac{dy}{dx} = g(x)\,h(y).$$

Then you separate and integrate both sides. Treating $\dfrac{dy}{dx}$ as a ratio is informal but valid here: gather all the y terms with dy and all the x terms with dx.

KEY

$$\frac{dy}{dx} = g(x)\,h(y) \quad\Longrightarrow\quad \int \frac{1}{h(y)}\,dy = \int g(x)\,dx$$

One arbitrary constant appears, giving the **general solution**, a family of curves. An **initial condition** fixes the constant and selects the **particular solution**.

EXAMPLE 17.1

Solve $\dfrac{dy}{dx} = 2xy^2$ given $y(0) = 1$.
Separate the variables and integrate:

$$\int \frac{1}{y^2}\,dy = \int 2x\,dx \implies -\frac{1}{y} = x^2 + C.$$

Apply $y(0) = 1$: $-1 = 0 + C \implies C = -1$. So $-\dfrac{1}{y} = x^2 - 1$, giving

$$y = \frac{1}{1 - x^2}.$$

Always apply the initial condition as early as possible; carrying C further than necessary invites errors.

EXAMPLE 17.2

A cup of coffee obeys $\dfrac{d\theta}{dt} = -k(\theta - 20)$ with $\theta(0) = 80$. Solve for $\theta(t)$.

Separate and integrate:

$$\int \frac{1}{\theta-20}\,d\theta = \int -k\,dt \implies \ln|\theta-20| = -kt + C.$$

Exponentiating, $\theta - 20 = Ae^{-kt}$. The condition $\theta(0) = 80$ gives $A = 60$, so

$$\theta(t) = 20 + 60e^{-kt}.$$

The temperature decays exponentially toward the room's 20 degrees, never quite reaching it: exactly the behaviour we observed.

Separation also handles the most famous population model in ecology. In the 1960s, once the cattle disease rinderpest was eliminated around the Serengeti, the wildebeest herd began to grow, and its growth obeyed the **logistic equation**: growth proportional both to the herd itself and to the room left below the plains' carrying capacity.

EXAMPLE 17.3

The Serengeti wildebeest population n (in millions) satisfies $\dfrac{dn}{dt} = kn(1.4-n)$, with $n(0) = 0.25$ in 1963. Solve for $n(t)$.

Separate, and split the left side with the partial fractions of Ch. 6:

$$\int \frac{dn}{n(1.4-n)} = \int k\,dt, \qquad \frac{1}{n(1.4-n)} = \frac{1}{1.4}\left(\frac{1}{n} + \frac{1}{1.4-n}\right).$$

Integrating,

$$\frac{1}{1.4}\ln\left|\frac{n}{1.4-n}\right| = kt + C \implies \frac{n}{1.4-n} = Be^{1.4kt}.$$

The initial condition gives $B = \dfrac{0.25}{1.15} \approx 0.217$, and solving for n:

$$n(t) = \frac{1.4}{1 + 4.6\,e^{-1.4kt}}.$$

This is the logistic S-curve: growth that is nearly exponential at first, then levels off at the carrying capacity of 1.4 million. The bracket $(1.4-n)$ is what slows it, since it shrinks towards zero as n approaches 1.4. The real herd behaved this way, climbing to about 1.3 million by 1977 and staying near that level since.

YOUR TURN 17.1 Separable Variables

1. Find the general solution of each equation.

(a) $\dfrac{dy}{dx} = \dfrac{x}{y}$ [2]

(b) $\dfrac{dy}{dx} = y$ [2]

(c) $\dfrac{dy}{dx} = \dfrac{1+y}{1+x}$ [3]

(d) $\dfrac{dy}{dx} = e^{x-y}$ [3]

2. Find the particular solution of each equation.

(a) $\dfrac{dy}{dx} = xy$ with $y(0) = 2$. [3]

(b) $\dfrac{dy}{dx} = 3x^2y$ with $y(0) = 1$. [3]

(c) $\dfrac{dy}{dx} = y\tan x$ with $y(0) = 4$, for $0 \le x < \frac{\pi}{2}$. [4]

(d) Explain why a general solution contains one arbitrary constant, and what the initial condition does to it. [2]

17.2 Homogeneous Differential Equations

Some equations are not separable as they stand, but become so after a substitution. A first-order equation is **homogeneous** if the right-hand side depends only on the ratio $\dfrac{y}{x}$. Such equations arise when a problem has no fixed scale, only directions. The standard example is a pursuit curve: a ferry that always aims at the dock while a current pushes it sideways. What matters at each moment is the direction from the ferry to the dock, and that direction is set by the ratio $\frac{y}{x}$ alone. You will solve the ferry in the challenge problems. The substitution $y = vx$ then separates the equation.

Since $y = vx$, the product rule gives $\dfrac{dy}{dx} = v + x\dfrac{dv}{dx}$. Substituting turns the equation into a separable one in v and x.

KEY For $\dfrac{dy}{dx} = f\left(\dfrac{y}{x}\right)$, substitute $y = vx$, so $\dfrac{dy}{dx} = v + x\dfrac{dv}{dx}$.

EXAMPLE 17.4

Solve $\dfrac{dy}{dx} = \dfrac{x+y}{x}$.

Rewrite the right side in terms of $\dfrac{y}{x}$: $\dfrac{x+y}{x} = 1 + \dfrac{y}{x}$. Substitute $y = vx$, $\dfrac{dy}{dx} = v + x\dfrac{dv}{dx}$:

$$v + x\frac{dv}{dx} = 1 + v \implies x\frac{dv}{dx} = 1.$$

This is separable:

$$\int dv = \int \frac{1}{x}\,dx \implies v = \ln|x| + C.$$

Finally replace $v = \dfrac{y}{x}$:

$$y = x(\ln|x| + C).$$

The single v on each side always cancels, and what is left is separable. That cancellation is what makes the substitution work.

YOUR TURN 17.2 Homogeneous Differential Equations

3. Use the substitution $y = vx$ to solve each equation.

(a) $\dfrac{dy}{dx} = \dfrac{2x+y}{x}$ [3]

(b) $\dfrac{dy}{dx} = \dfrac{y-x}{x}$ [3]

(c) $\dfrac{dy}{dx} = \dfrac{x+2y}{x}$ [4]

(d) $\dfrac{dy}{dx} = \dfrac{y}{x}$ for $x > 0$. [2]

4.

(a) Show that $\dfrac{dy}{dx} = \dfrac{x^2+y^2}{xy}$ is homogeneous, by writing the right-hand side in terms of $\dfrac{y}{x}$. [3]

(b) Hence find its general solution. [5]

17.3 The Integrating Factor

A **linear** first-order differential equation has the form

$$\frac{dy}{dx} + P(x)\,y = Q(x).$$

It is usually not separable. The method is to multiply through by a carefully chosen function, the **integrating factor**, which turns the left side into the derivative of a single product.

KEY · IB For $\dfrac{dy}{dx} + P(x)y = Q(x)$, the integrating factor is

$$I = e^{\int P(x)\,dx}.$$

After multiplying by I, the left side becomes exactly $\dfrac{d}{dx}(Iy)$, by the product rule run backwards. You then integrate both sides directly.

EXAMPLE 17.5

Solve $\dfrac{dy}{dx} + \dfrac{1}{x}y = x$ for $x > 0$.

Here $P(x) = \dfrac{1}{x}$, so the integrating factor is

$$I = e^{\int \frac{1}{x}\,dx} = e^{\ln x} = x.$$

Multiply through by x:

$$x\frac{dy}{dx}+y=x^2 \implies \frac{d}{dx}(xy)=x^2.$$

The left side has collapsed to a single derivative. Integrate both sides:

$$xy=\frac{x^3}{3}+C \implies y=\frac{x^2}{3}+\frac{C}{x}.$$

The whole method exists to create that $\frac{d}{dx}(xy)$ on the left; everything after is ordinary integration.

EXAMPLE 17.6

Solve $\frac{dy}{dx}+y=e^x$.
$P(x)=1$, so $I=e^{\int 1\,dx}=e^x$. Multiply through:

$$\frac{d}{dx}(e^x y)=e^x\cdot e^x=e^{2x}.$$

Integrate: $e^x y=\frac{1}{2}e^{2x}+C$, hence $y=\frac{1}{2}e^x+Ce^{-x}$.

Where do such equations come from? The classic source is a **mixing problem**: a tank where liquid flows in and out, and the balance of inflow and outflow writes itself as a linear equation. When the volume itself changes, the equation is linear but not separable, so the integrating factor is the method to reach for.

EXAMPLE 17.7

A 100-litre vat in a brewery holds 5 kg of dissolved sugar. Syrup carrying 0.2 kg of sugar per litre flows in at 1 litre per minute, the vat is kept well stirred, and the mixture drains at 2 litres per minute. Find the sugar $S(t)$ in the vat at time t minutes.
The volume shrinks: $V(t)=100-t$. Sugar arrives at $0.2\times 1=0.2$ kg min^{-1} and leaves at concentration times outflow, $\frac{S}{100-t}\times 2$. So

$$\frac{dS}{dt}=0.2-\frac{2S}{100-t} \implies \frac{dS}{dt}+\frac{2}{100-t}S=0.2.$$

The integrating factor is

$$I=e^{\int \frac{2}{100-t}\,dt}=e^{-2\ln(100-t)}=(100-t)^{-2}.$$

Multiplying through collapses the left side:

$$\frac{d}{dt}\Big(S\,(100-t)^{-2}\Big)=0.2\,(100-t)^{-2} \implies S\,(100-t)^{-2}=\frac{0.2}{100-t}+C.$$

So $S=0.2(100-t)+C(100-t)^2$, and $S(0)=5$ gives $C=-0.0015$:

$$S(t)=0.2(100-t)-0.0015(100-t)^2.$$

Sense-check the endpoints: $S(0)=5$, and $S(100)=0$, an empty vat holds no sugar. In be-

> tween, the sugar first *rises*, to about 6.7 kg near $t = 33$, before the draining takes over. That rise is not obvious from the description, and it is the equation rather than intuition that reveals it.

Choosing a method

Three exact methods are now available, and a question rarely says which to use. The form of the equation decides it, and the test takes one line.

What the equation looks like	Method	Example
The variables can be split, $\frac{dy}{dx} = g(x)h(y)$	separate and integrate	$\frac{dy}{dx} = 2xy^2$
The right side depends only on $\frac{y}{x}$	substitute $y = vx$	$\frac{dy}{dx} = \frac{x+y}{x}$
Linear in y: $\frac{dy}{dx} + P(x)y = Q(x)$	integrating factor $e^{\int P\,dx}$	$\frac{dy}{dx} + \frac{1}{x}y = x$
None of the above	Euler's method, numerically	$\frac{dv}{dt} = 9.8 - 0.1v^2$

Try them in that order, because each test is quicker than the one after it. Note that the first two overlap: an equation can be both separable and homogeneous, in which case separating directly is the shorter route. Note also that a linear equation is separable only when $Q(x)$ is zero, which is why the integrating factor exists at all.

YOUR TURN 17.3 The Integrating Factor

5. Solve each linear equation.

(a) $\frac{dy}{dx} + y = 4$ [3]

(b) $\frac{dy}{dx} + 2y = 6$ [3]

(c) $\frac{dy}{dx} + 2y = e^x$ [3]

(d) $\frac{dy}{dx} - y = e^{2x}$ [3]

6.

(a) Write down the integrating factor for $\frac{dy}{dx} + \frac{3}{x}y = x^2$, for $x > 0$. [2]

(b) Solve $\frac{dy}{dx} + \frac{2}{x}y = x$ for $x > 0$. [3]

(c) Solve $\dfrac{dy}{dx} + \dfrac{1}{x}y = x$ with $y(1) = 1$, for $x > 0$. [4]

(d) Explain what the integrating factor is chosen to achieve on the left-hand side. [2]

17.4 Euler's Method

Not every differential equation can be solved exactly, and most of the equations that describe the real world cannot. When no exact solution exists, the solution can still be approximated numerically. **Euler's method** does this by walking along the curve in short straight steps, taking the gradient of each step from the differential equation itself.

The idea is the tangent line of Ch. 14. Near a known point the curve is close to its tangent, so moving a small distance h along the tangent lands close to the curve:

$$y \approx y_0 + \underbrace{f(x_0, y_0)}_{\text{gradient}} \times \underbrace{h}_{\text{step}} .$$

The differential equation supplies the gradient at whatever point you have reached, so the same move can be repeated from the new point, and again from the one after that.

KEY · IB

$$y_{n+1} = y_n + h\,f(x_n, y_n), \qquad x_{n+1} = x_n + h$$

Here h is the fixed **step length**. Each step is exact only at its starting point and drifts from the curve thereafter, so shorter steps stay closer, at the cost of needing more of them.

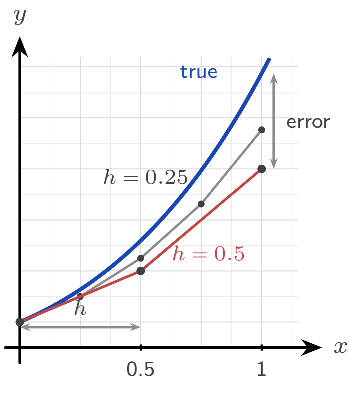

Figure 17.1 Euler's method on $\frac{dy}{dx} = x + y$ with $y(0) = 1$. Each straight step leaves the curve, which bends upward away from it, so the estimate falls short and the gap at $x = 1$ is the error. Halving the step length roughly halves that gap.

The figure shows two things at once. Each step is a straight line, so it leaves a curve that bends away from it, and the gap at the far end is the *error*. The gray path takes steps half as long and stays about twice as close to the curve. Shorter steps give more accuracy, at the cost of more of them.

EXAMPLE 17.8

Given $\dfrac{dy}{dx} = x + y$ with $y(0) = 1$, use Euler's method with $h = 0.1$ to estimate $y(0.3)$.
Apply $y_{n+1} = y_n + h(x_n + y_n)$ three times. A table is the clearest way to organise the work, and it is what an examiner expects to see.

n	x_n	y_n	$f = x_n + y_n$	$h\,f$
0	0	1	1	0.1
1	0.1	1.1	1.2	0.12
2	0.2	1.22	1.42	0.142
3	0.3	1.362		

Each y_n is the previous one plus the $h\,f$ on the line above. So $y(0.3) \approx 1.362$, against an exact value of about 1.3997.
Euler has undershot, and the reason is visible in the figure: this curve is concave up, so every tangent lies below it and each step falls short. That is the general rule. A concave-up solution is underestimated and a concave-down one is over-estimated, so the sign of $\frac{d^2y}{dx^2}$ tells you which way your answer is wrong before you check it.

The method matters most when no exact solution is available. Add air resistance to a falling body and the equation becomes nonlinear, with no solution among the functions of this course, but Euler's method still produces a table of values. This is, in miniature, how weather forecasts are computed: equations too hard to solve exactly, stepped forward numerically, millions of times.

EXAMPLE 17.9

A skydiver falls from rest, and air resistance gives $\dfrac{dv}{dt} = 9.8 – 0.1v^2$ (in m s^{-1} and seconds). Use Euler's method with $h = 0.5$ to estimate the speed after 1.5 seconds, and compare with the terminal velocity.
Apply $v_{n+1} = v_n + 0.5(9.8 - 0.1v_n^2)$ three times from $v_0 = 0$:

$$
\begin{aligned}
v_1 &= 0 + 0.5(9.8) = 4.9 \\
v_2 &= 4.9 + 0.5(9.8 - 0.1(4.9)^2) = 4.9 + 0.5(7.399) = 8.60 \\
v_3 &= 8.60 + 0.5(9.8 - 0.1(8.60)^2) = 8.60 + 0.5(2.404) = 9.80
\end{aligned}
$$

So $v(1.5) \approx 9.80$ m s^{-1}. The terminal velocity is where acceleration dies: $9.8 - 0.1v^2 = 0 \implies v = \sqrt{98} \approx 9.90$ m s^{-1}. After only three steps the estimate is already within 1% of the terminal speed. The physics is visible in the arithmetic: each bracket $9.8 – 0.1v_n^2$ is the net force per unit mass, and it shrinks towards zero as the speed builds.

YOUR TURN 17.4 Euler's Method

7. Use Euler's method for each of the following.

(a) $\dfrac{dy}{dx} = 2x$, $y(0) = 1$; use $h = 0.5$ to estimate $y(1)$. [3]

(b) $\dfrac{dy}{dx} = y$, $y(0) = 1$; use $h = 0.1$ to estimate $y(0.2)$. [3]

(c) $\dfrac{dy}{dx} = x + y$, $y(0) = 1$; use $h = 0.2$ to estimate $y(0.4)$. [3]

(d) $\dfrac{dy}{dx} = xy$, $y(1) = 1$; use $h = 0.1$ to estimate $y(1.2)$. [3]

8.

(a) For $\dfrac{dy}{dx} = xy$ with $y(1) = 2$, take one step with $h = 0.2$ to estimate $y(1.2)$. [2]

(b) For $\dfrac{dy}{dx} = x + y$ with $y(0) = 0$, take two steps with $h = 0.5$ to estimate $y(1)$. [3]

(c) Solve $\dfrac{dy}{dx} = y$ with $y(0) = 1$ exactly, and compare the exact $y(0.2)$ with your answer to question 7 part (b). [3]

(d) State whether Euler's method over- or under-estimates the solution of part (c), and explain why in terms of the shape of the curve. [3]

17.5 Maclaurin Series

Here is a different idea, and a powerful one: many functions can be written as an infinite polynomial.

It is worth knowing why anyone would want that. A polynomial is built from adding and multiplying, and those are the only two operations a machine can actually perform. Nothing in a calculator knows what $\sin x$ *is*. When you press the sine key it evaluates a polynomial, because a polynomial is something arithmetic can reach. Every function outside that small arithmetic world has to be turned into one before it can be computed, and a Maclaurin series is how that is done.

The idea is to match the function at a single point, as closely as possible. Near $x = 0$, look for a polynomial that shares the function's value, then its gradient, then its concavity, then every higher derivative at that point. The more derivatives agree, the longer the polynomial stays with the curve as you move away from 0. This is the **Maclaurin series**.

The coefficients are not chosen; they are forced. Suppose the series exists and write it as

$$f(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \cdots$$

Putting $x = 0$ kills every term but the first, so $a_0 = f(0)$. Differentiating gives $f'(x) =$

$a_1 + 2a_2x + 3a_3x^2 + \cdots$, and $x = 0$ now leaves $a_1 = f'(0)$. Differentiating again gives $f''(0) = 2a_2$, and once more $f'''(0) = 6a_3 = 3!\,a_3$. Each round of differentiating strips one term and multiplies the next by its own index, so in general $f^{(n)}(0) = n!\,a_n$.

KEY · IB

$$f(x) = f(0) + f'(0)\,x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \cdots$$

EXAMPLE 17.10

Find the Maclaurin series for $f(x) = e^x$.

Every derivative of e^x is e^x, and $e^0 = 1$, so $f^{(n)}(0) = 1$ for all n:

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$$

This series can be taken as the definition of e^x, and it computes the function to any accuracy you want. It also explains why e^x is its own derivative: differentiate the series term by term and you get the same series back.

The five standard series below are provided in the formula booklet. Each is exact, and truncating it gives a polynomial approximation that is excellent near $x = 0$.

KEY · IB

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots \qquad \sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots \qquad \ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots$$

$$\arctan x = x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots$$

The syllabus adds one more family: $(1+x)^p$ for any rational p, whose expansion $1 + px + \frac{p(p-1)}{2!}x^2 + \cdots$ is exactly the extended binomial theorem of Ch. 2 (booklet 1.10). With $p = \frac{1}{2}$, for instance, $\sqrt{1+x} \approx 1 + \frac{x}{2} - \frac{x^2}{8}$, which gives $\sqrt{1.1} \approx 1.04875$ without a calculator. The true value is 1.04881.

The figure below shows the first three odd-degree truncations of the sine series: the line $y = x$, the cubic, and the quintic. Each new term extends the interval over which the polynomial and the curve agree.

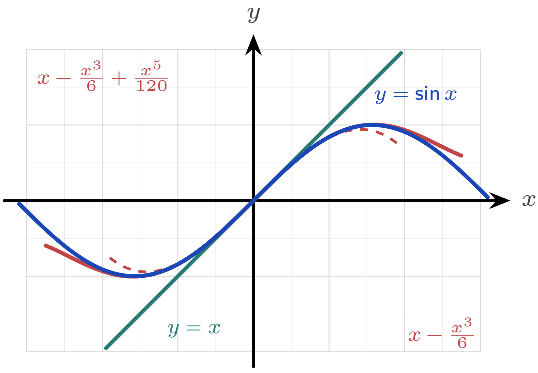

Figure 17.2 The first three odd-degree truncations of the sine series against $y = \sin x$. Each extra term widens the interval on which the polynomial and the curve agree, but every truncation eventually leaves the curve.

The same convergence can be watched numerically. Below are the successive truncations of the sine series at $x = 1$, against the true value $\sin 1 = 0.841471$.

terms up to	x	x^3	x^5	x^7
estimate	1	0.833333	0.841667	0.841468
error	0.159	0.0081	0.0002	0.0000027

Each new term cuts the error by a factor of tens, then hundreds, then thousands. That is the trade the series offers: a polynomial is simple enough to compute, and taking more terms buys as much accuracy as you are willing to pay for.

Building new series

New series are rarely built from the definition. They are made from the standard five by four operations, all of which the syllabus names.

- **Substitute.** Replacing x with $-x^2$ in the series for e^x gives

$$e^{-x^2} = 1 - x^2 + \frac{x^4}{2!} - \frac{x^6}{3!} + \cdots$$

- **Differentiate.** Differentiating the sine series term by term produces exactly the cosine series: the identity $\frac{d}{dx}\sin x = \cos x$ in polynomial form.
- **Integrate.** The geometric series of Ch. 2 gives $\dfrac{1}{1+x^2} = 1 - x^2 + x^4 - \cdots$ for $|x| < 1$, and integrating term by term yields $\arctan x = x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots$. The booklet's arctangent series is simply the geometric series integrated.
- **Multiply.** For a product, expand each factor and collect powers up to the degree you need.

> **EXAMPLE 17.11**
>
> Find the Maclaurin series of $e^x \sin x$ up to the term in x^3.
> Multiply the two standard series, keeping only terms of degree ≤ 3:
>
> $$\left(1 + x + \frac{x^2}{2} + \frac{x^3}{6}\right)\left(x - \frac{x^3}{6}\right) = x + x^2 + \left(\frac{1}{2} - \frac{1}{6}\right)x^3 + \cdots = x + x^2 + \frac{x^3}{3} + \cdots$$
>
> The work is bookkeeping. Decide the target degree first, and discard anything above it before multiplying rather than after.

Small angles and series from differential equations

Truncating at the first term or two gives the **small-angle approximations** used throughout physics. For θ near 0, in radians,

$$\sin\theta \approx \theta, \qquad \cos\theta \approx 1 - \frac{\theta^2}{2}, \qquad \tan\theta \approx \theta.$$

These are not conveniences invented by physicists; they are the first one or two terms of the series, with the rest discarded. The pendulum formula depends on the first of them. For small swings the restoring force involves $\sin\theta$, and replacing $\sin\theta$ by θ is exactly what turns the equation of motion into simple harmonic motion, which can then be solved.

The series also prices the approximation. The first term discarded is $\frac{\theta^3}{6}$, so that is roughly the error, and the relative error stays under 1% for swings up to about 14°. Beyond that the pendulum's real period drifts away from the textbook formula, by an amount large enough to measure.

A Maclaurin series can also be built directly *from a differential equation*, by differentiating the equation itself to obtain the derivatives at 0. This joins the two halves of the chapter.

EXAMPLE 17.12

A function satisfies $\dfrac{dy}{dx} = 1 + y^2$ with $y(0) = 0$. Find the Maclaurin series of y up to the term in x^3.

Take the derivatives at 0 from the equation itself. First, $y'(0) = 1+0 = 1$. Differentiating the equation,

$$y'' = 2y\,y' \implies y''(0) = 0, \qquad y''' = 2(y')^2 + 2y\,y'' \implies y'''(0) = 2.$$

Feed these into the Maclaurin formula:

$$y = 0 + x + \frac{0}{2!}x^2 + \frac{2}{3!}x^3 + \cdots = x + \frac{x^3}{3} + \cdots$$

You may recognise the answer: $y = \tan x$ solves this equation, since $\frac{d}{dx}\tan x = \sec^2 x = 1 + \tan^2 x$, and these are indeed the first terms of its series. The equation was never solved. Each derivative at 0 was read off from the equation itself, one at a time, and that was enough to build the series.

TIP *Maclaurin series suggest several good IA topics. Two worth exploring: Madhava's series* $\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \cdots$, *which comes from setting* $x = 1$ *in the arctangent series, and why it converges so slowly that a better choice such as* $\arctan \frac{1}{\sqrt{3}}$ *reaches the same accuracy in a fraction of the terms; or how a calculator really evaluates* $\sin x$, *by reducing the angle to a small interval first and then using a short truncated series.*

EXAMPLE 17.13

Use Maclaurin series to evaluate $\lim\limits_{x\to 0} \dfrac{e^x - 1 - x}{x^2}$.

Expand the numerator with the series for e^x:

$$e^x - 1 - x = \left(1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \cdots\right) - 1 - x = \frac{x^2}{2} + \frac{x^3}{6} + \cdots$$

Divide by x^2:

$$\frac{e^x - 1 - x}{x^2} = \frac{1}{2} + \frac{x}{6} + \cdots \xrightarrow{x\to 0} \frac{1}{2}.$$

The series settles an indeterminate limit as neatly as L'Hôpital's rule, and it shows *why* the answer is $\frac{1}{2}$: that number is the leading coefficient.

YOUR TURN 17.5 Maclaurin Series

9. Find the first three nonzero terms of the Maclaurin series for each function.

(a) e^{2x} [2] (b) $\cos x$ [2]

(c) $\ln(1 + 2x)$ [2] (d) e^{-x^2} [3]

10. Find the first two nonzero terms of the Maclaurin series for each function.

(a) $\sin 2x$ [2] (b) $\sin(x^2)$ [2]

(c) $\arctan 2x$ [2] (d) $\sqrt{1+x}$ [3]

11.

(a) Use the first three terms of the series for e^x to estimate $e^{0.1}$. [2]

(b) Use the first three terms of the cosine series to estimate $\cos 0.2$ to four decimal places. [3]

(c) Compare each estimate with the true value, and state which is the more accurate. Explain why. [3]

(d) Find the first three nonzero terms of the Maclaurin series for $e^x \cos x$. [4]

12. A function satisfies $\dfrac{dy}{dx} = x + y$ with $y(0) = 1$.

(a) Write down $y'(0)$. [1]

(b) By differentiating the equation, find $y''(0)$ and $y'''(0)$. [4]

(c) Hence find the Maclaurin series for y up to the term in x^3. [3]

(d) Use the series to estimate $y(0.3)$, and compare it with the Euler estimate 1.362 from the notes. [3]

Chapter 17 · Differential Equations

Reflections

RW Differential equations are the language in which physics is written. Newton's second law $F = ma$ is a differential equation for position; radioactive decay, population growth, electrical circuits, epidemics, and planetary orbits are all governed by them. To predict a system's future is, almost always, to solve a differential equation, exactly when you can and numerically (by methods like Euler's) when you cannot.

TOK Euler's method rests on a striking idea: knowing the rule for change at every point, plus a single starting state, determines the entire future of a system. This is mathematical determinism. Yet the same equations can be so sensitive to their starting values that long-term prediction becomes impossible in practice, the phenomenon of chaos. Determined but unpredictable: what does that tell us about the limits of knowledge?

IM Both halves of this chapter have a long international lineage.

The series came first from India. Mathematicians of the Kerala school, notably Madhava in the 14th century, found the expansions of $\sin x$, $\cos x$ and $\arctan x$ roughly three centuries before Newton, Leibniz and Maclaurin. The arctangent series, usually named after Gregory and Leibniz, was known in Kerala generations earlier.

Numerical methods have a similar story in the Islamic world. In the 15th century Jamshīd al-Kāshī, working at the observatory in Samarkand, needed $\sin 1°$ to extraordinary precision for his astronomical tables. No formula gave it, so he wrote down an equation the value satisfied and solved it by repeated approximation, each round feeding its answer into the next, reaching sixteen correct decimal places. That is the same bargain Euler's method strikes: when no exact answer is available, iterate towards one and control the error. Two centuries earlier Sharaf al-Dīn al-Ṭūsī had studied when a cubic equation has solutions by examining where its maximum lies, an argument that reads today like the use of a derivative.

Mathematical discovery is rarely the work of one place or one name.

TOK A truncated series is deliberately wrong. It is chosen because it is simple enough to compute, and every extra term bought accuracy at the cost of that simplicity: four terms of the sine series give $\sin 1$ correct to five decimal places, and one term gives it to none.

Is there always a trade-off between accuracy and simplicity? And when a model is knowingly approximate, as every model in this chapter is, what makes it good enough to act on?

EXT The five standard series behave differently at the edges. The series for e^x, $\sin x$ and $\cos x$ converge for every x, but $\arctan x$ only for $|x| \leq 1$ and $\ln(1+x)$ only for $-1 < x \leq 1$, while the geometric series they are built from needs $|x| < 1$. Each series has a *radius of convergence*, and outside it the polynomial does not merely lose accuracy: it diverges completely. Finding that radius, and explaining why $\ln(1+x)$ has one at all when e^x does not, is the beginning of complex analysis, where the answer turns out to depend on where the function misbehaves in the complex plane. See Ch. 18.

EXT Maclaurin series turn calculus into algebra: integrals with no elementary antiderivative, like $\int e^{-x^2}\,dx$, can be integrated term by term once the function is a series. Series also solve differential equations that no other method in this course can reach, by assuming the answer is a power series and matching coefficients. The infinite polynomial is one of the most far-reaching ideas in all of analysis.

End-of-Chapter Problems

1. A population grows so that $\dfrac{dP}{dt} = 0.2P$, with $P = 500$ when $t = 0$.

(a) Solve the differential equation for $P(t)$. [4]

(b) Find the population when $t = 10$. [2]

2. A body cools according to $\dfrac{d\theta}{dt} = -0.1(\theta - 25)$, with $\theta(0) = 90$ (in °C).

(a) Solve for $\theta(t)$. [4]

(b) Find the temperature after 10 minutes. [2]

(c) State the temperature the body approaches in the long run. [1]

3. Consider $\dfrac{dy}{dx} = \dfrac{x^2 + y^2}{xy}$ for $x > 0$.

(a) Use the substitution $y = vx$ to show that $xv\dfrac{dv}{dx} = 1$. [4]

(b) Hence find the general solution. [3]

4. Consider $\dfrac{dy}{dx} + \dfrac{1}{x}y = 3x$ for $x > 0$.

(a) Find the integrating factor. [2]

(b) Solve the equation given $y(1) = 2$. [4]

5. Consider $\dfrac{dy}{dx} = x + y$, with $y(0) = 1$.

(a) Use Euler's method with $h = 0.25$ to estimate $y(0.5)$, showing each step. [4]

(b) Solve the equation exactly (integrating factor) and find the true $y(0.5)$. [4]

(c) Comment on the size and direction of the error. [2]

6. Let $f(x) = \cos x$.

(a) Find $f(0), f'(0), f''(0), f'''(0), f^{(4)}(0)$. [3]

(b) Hence write the Maclaurin series of $\cos x$ up to the term in x^4. [2]

(c) Use it to estimate $\cos(0.2)$. [2]

7.

(a) Write down the Maclaurin series for $\sin x$ up to the term in x^5. [2]

(b) Hence evaluate $\lim\limits_{x \to 0} \dfrac{x - \sin x}{x^3}$. [3]

8.

(a) Write down the Maclaurin series for e^x up to the term in x^3. [1]

(b) By substituting $-x^2$, find the series for e^{-x^2} up to x^4. [2]

(c) Hence estimate $\int_0^1 e^{-x^2}\,dx$ using terms up to x^4. [4]

9. A bowl of miso soup at $90°$C is left in a $25°$C room and obeys $\frac{d\theta}{dt} = -k(\theta - 25)$. After 10 minutes it has cooled to $65°$C.

(a) Solve the differential equation, expressing θ in terms of k and t. [4]

(b) Find k. [2]

(c) Find how long, from the start, the soup takes to reach $40°$C. [2]

10. Consider $\frac{dy}{dx} = \frac{y^2 + xy}{x^2}$ for $x > 0$, with $y(1) = 1$.

(a) Use the substitution $y = vx$ to show that $x\frac{dv}{dx} = v^2$. [3]

(b) Hence solve for y in terms of x. [4]

11. Solve each linear equation.

(a) $\frac{dy}{dx} - \frac{y}{x} = x^2$ for $x > 0$, given $y(1) = 1$. [6]

(b) $\frac{dy}{dx} + 2y = e^{-x}$ with $y(0) = 0$. [6]

12. Solve $\frac{dy}{dx} + (\tan x)\, y = \sec x$ for $-\frac{\pi}{2} < x < \frac{\pi}{2}$, given $y(0) = 2$. [7]

13. A fish population n (in thousands) in a lake satisfies the logistic equation $\frac{dn}{dt} = 0.4n(2-n)$, with $n(0) = 0.5$.

(a) Using partial fractions, show that the solution is $n = \frac{2}{1 + 3e^{-0.8t}}$. [6]

(b) Find the time at which the population reaches 1500 fish. [3]

14. Consider $\frac{dy}{dx} = y - x^2$ with $y(0) = 1$.

(a) Use Euler's method with $h = 0.25$ to estimate $y(0.5)$. [5]

(b) The exact solution is concave up on $0 \le x \le 0.5$. State, with a reason, whether your estimate is above or below the true value. [2]

15. A skydiver falls from rest with $\frac{dv}{dt} = 9.8 - 0.2v^2$.

(a) Use Euler's method with $h = 0.5$ to estimate $v(1.5)$. [4]

(b) Find the terminal velocity, and comment on the behaviour of the Euler estimates around it. [3]

16. Find the Maclaurin series of $\ln(1+2x)$ up to the term in x^3

(a) by substitution into a standard series; [3]
(b) by computing derivatives from the definition. [3]

17.

(a) Find the Maclaurin series of $e^{-x}\sin x$ up to the term in x^3. [6]
(b) Use your series to estimate $e^{-0.1}\sin 0.1$, and compare it with the value from your calculator. [3]

18. By setting $x = 1$ in the Maclaurin series for $\arctan x$:

(a) write down a series for $\frac{\pi}{4}$; [2]
(b) estimate π using the first four terms; [2]
(c) comment on the rate of convergence. [1]

19. Use Maclaurin series to evaluate $\lim_{x\to 0}\frac{x-\sin x}{x^3}$. [5]

20. A function satisfies $\frac{dy}{dx} = x + y^2$ with $y(0) = 1$. Find the Maclaurin series of y up to the term in x^3. [7]

21. A drug is infused into the bloodstream at a constant rate r mg per hour, and the body removes it at a rate proportional to the concentration C, so that

$$\frac{dC}{dt} + kC = r,$$

where t is measured in hours and $C(0) = 0$.

(a) Explain why this equation is linear, and write down its integrating factor. [3]
(b) Solve the equation to show that $C = \frac{r}{k}(1 - e^{-kt})$. [4]
(c) A patient is given $r = 6$ and has $k = 0.3$. Find the concentration after 4 hours. [2]
(d) Find the concentration the patient approaches in the long run, and explain what this means for the treatment. [3]
(e) Find how long it takes to reach 90% of that level. [2]
(f) Once that level is reached the infusion is stopped, so that $\frac{dC}{dt} = -kC$. Solve this equation and find how long the concentration then takes to halve. [3]

22. In this question you will build a series for logarithms that converges quickly.

(a) Find the Maclaurin series of $\ln(1+x)$ up to the term in x^4. [4]
(b) Write down the Maclaurin series of $\ln(1-x)$ up to the term in x^4. [2]
(c) Hence show that $\ln\left(\frac{1+x}{1-x}\right) = 2\left(x + \frac{x^3}{3} + \frac{x^5}{5} + \cdots\right)$. [3]

(d) By choosing a suitable value of x, use the three terms shown to estimate $\ln 3$. [3]

(e) The series for $\ln(1+x)$ is valid only for $-1 < x \leq 1$. State the value of x that would be needed to reach $\ln 3$ from that series alone, and explain why the series in part (c) is the more useful one here. [2]

23. Water drains from a tank through a hole in its base. The depth h metres at time t minutes satisfies

$$\frac{dh}{dt} = -k\sqrt{h},$$

where k is a positive constant.

(a) Separate the variables to show that $2\sqrt{h} = c - kt$ for some constant c. [3]

(b) Given that the depth is h_0 when $t = 0$, show that $h = \left(\sqrt{h_0} - \frac{1}{2}kt\right)^2$. [2]

(c) A tank starts with $h_0 = 1.44$ and has $k = 0.2$. Find how long it takes to empty. [3]

(d) Find the depth after 6 minutes. [2]

(e) Use Euler's method with steps of 3 minutes to estimate the depth after 6 minutes, and find the error in this estimate. [4]

(f) Describe the shape of the graph of h against t, and use it to explain why the Euler estimate lies below the exact value. [2]

Challenge Questions

24. How fast does coffee cool? The chapter opened with $\dfrac{d\theta}{dt} = -k(\theta - 20)$, where θ is the temperature in degrees and t the time in minutes. A cup is poured at 80°, so $\theta(0) = 80$.

(a) Solve the equation to show that $\theta = 20 + 60e^{-kt}$. [3]

(b) The coffee has cooled to 60° after 10 minutes. Find k exactly, and to three significant figures. [3]

(c) Find the temperature after 20 minutes. [2]

(d) Show that the time taken to reach a temperature θ is $t = \dfrac{1}{k}\ln\left(\dfrac{60}{\theta - 20}\right)$, and explain why the coffee never reaches 20°. [4]

(e) A second cup is poured at 90° in the same room. Show that it takes exactly 10 minutes to cool from 80° to 60°, the same as the first cup. [3]

25. How wrong is Euler? Take $\dfrac{dy}{dx} = y$ with $y(0) = 1$, whose exact solution is $y = e^x$, so that $y(1) = e$.

(a) Show that one Euler step of length h gives $y_{n+1} = (1+h)y_n$, and hence that after n steps $y_n = (1+h)^n$. [3]

(b) Taking $h = \dfrac{1}{n}$, deduce that Euler's estimate of $y(1)$ is $\left(1 + \dfrac{1}{n}\right)^n$. [2]

(c) Evaluate this estimate for $n = 2, 4, 8, 16$, and tabulate the error in each case. [4]

(d) Describe what happens to the error each time h is halved, and state what this says about the accuracy of the method. [3]

(e) Write down the limit of $\left(1+\dfrac{1}{n}\right)^n$ as $n \to \infty$, and explain why Euler's method had to give this answer. [3]

26. The ferry and the current. A ferry sets off from $(1, 0)$ toward a dock at the origin, always aiming straight at the dock, while a river current pushes it in the positive y-direction at a fraction k of the ferry's own speed. Its path satisfies

$$\frac{dy}{dx} = \frac{y - k\sqrt{x^2+y^2}}{x}.$$

(a) Show the equation is homogeneous, and that $y = vx$ leads to $x\dfrac{dv}{dx} = -k\sqrt{1+v^2}$. [3]

(b) Given that $\displaystyle\int \frac{dv}{\sqrt{1+v^2}} = \ln\left(v+\sqrt{1+v^2}\right) + C$, show that $v + \sqrt{1+v^2} = x^{-k}$. [3]

(c) Hence show that $y = \frac{1}{2}\left(x^{1-k} - x^{1+k}\right)$. [4]

(d) Taking $k = \frac{1}{2}$, find where the ferry lands. [2]

(e) Show that the ferry reaches the dock only when $k < 1$, and describe what happens when $k = 1$ and when $k > 1$. [4]